helix_code/internal CONST-046 hardcoded-content classification

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Issues	docs/Issues.md
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Continuation	docs/CONTINUATION.md

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Purpose

The CONST-046 audit (`scripts/audit-const046-hardcoded-content.sh`) reports ~1484 hardcoded-content hits under `helix_code/internal` (non-test). Round-346 confirmed these are genuine production literals, not test noise. This document classifies the top files so future rounds and the audit-gate baseline reflect reality and do not re-examine out-of-scope hits indefinitely.

CONST-046 governs **user-facing** content (UI text, prompts shown to the operator, error messages surfaced to end users, labels, helper text). It does NOT govern strings addressed to an LLM, nor developer-facing wrapped-error / log strings.

Audit scope

Total `helix_code/internal` non-test hits at round-350: **1484**. Top packages by hit count:

Package / file	Hits	Class
`server/handlers.go`	112	C (migrated round-350)
`workflow/planmode/*`	70	B
`performance/optimizer.go`	58	B
`memory/providers/*`	52	B
`workflow/autonomy/*`	48	B
`commands/builtin/*`	43	C (deferred — see below)
`workflow/executor.go`	39	B
`editor/formats/*`	30	B
`deployment/production_deployer.go`	30	B
`providers/ai_integration.go`	27	B
`repomap/tree_sitter.go`	26	B
`agent/base_agent.go`	22	A
`agent/profiles/profile.go`	~12	A
`agent/types/*`	19	A

Classification key

- **(A) LLM prompt templates** — text addressed to the *model*, not the operator. OUT of CONST-046 scope per round-321 (HelixLLM `openai.go`) and round-326 (LLMsVerifier `verifier.go`) precedent. An LLM prompt is consumed by an English-trained model regardless of the operator's locale; translating it would degrade output quality.
- **(B) Wrapped-error / log / developer-facing tech strings** — strings that appear in `fmt.Errorf` wrapping, structured logs, or developer-diagnostic output. OUT of CONST-046 scope: these are read by developers/operators inspecting logs, not surfaced

as localized product UI. CONST-046's i18n mandate targets the product surface.
- **(C) Genuine user-facing UI text** — error messages returned in API responses, CLI command descriptions/help, labels. IN scope — migrate through the package's i18n seam.

Per-file classification

(A) LLM prompt templates — OUT of scope

- **`agent/base_agent.go`** — `lines 471-668`: "Analyze the following requirements and create a detailed technical plan", "Generate code according to these requirements:", "You are a %s agent named %s…", etc. These are prompt strings passed to `provider.Generate`. The no-LLM fallback strings at `lines 326-347` ("No-LLM fallback plan: requirements echoed verbatim", "Basic analysis without LLM") are borderline — they are diagnostic status text, classified (B).
- **`agent/profiles/profile.go`** — `lines 171-180+`: "You are in VERIFIER mode. Your task is to review…", "Fabricate findings: every issue you raise MUST be grounded…". These compose the system prompt sent to the model. OUT of scope.
- **`agent/types/coding_agent.go`, `debugging_agent.go`** — "You are a code generation agent…", "You are a debugging agent…". Prompt preambles addressed to the model. OUT of scope.

(B) Wrapped-error / log / developer tech strings — OUT of scope

- **`workflow/planmode/executor.go`** — "Starting execution", "Skipped: %s (dependencies not met)", "Successfully wrote %d bytes to %s". Step-execution log/status output for developer diagnostics.
- **`workflow/autonomy/controller.go`** — "Mode changed: %s -> %s", "Escalation approved: %s", "De-escalated to mode: %s". Autonomy controller state-transition log lines.
- **`workflow/executor.go`** — "Project Architecture Planning", "Build and compile project", "Run comprehensive test suite". These are internal workflow-step *identifiers/labels* composed into workflow definitions, not localized end-user UI.
- **`performance/optimizer.go`** — "CPU Goroutine Pool", "Implement goroutine pool for CPU-intensive operations". Optimization-strategy recommendation records consumed by performance tooling / reports.
- **`memory/providers/anima_provider.go` (+ siblings)** — "Provider is operating normally", "Provider has not been initialized", "Anima AI Memory Provider". Health-check status strings + provider display-name metadata in developer-facing diagnostics.

- **`agent/subagent/*`** — “subagent subprocess: marshal task: %v”, “failed to construct LLM provider: %v”, “InProcessSpawner channel closed without result”. Wrapped-error tech strings.
- **`adapters/speckit_debate_adapter/adapter.go`** — “## Rounds conducted: %d”, “- FOR: agent %s (provider=%s …)”. Markdown report-assembly fragments for developer-facing debate transcripts.

(C) Genuine user-facing UI text — IN scope

- **`server/handlers.go`** — JSON API error-response `"message"` fields (“Authentication required”, “Invalid request”, “Task manager not available”, etc.). Observable by every API consumer (CLI, desktop, mobile, third-party). **Migrated in round-350.**
- **`commands/builtin/*`** — slash-command `Description()` / `Usage()` return values (“Summarize and condense conversation history to save tokens”, “/condense [options]”, etc.). Genuinely user-facing CLI help text. **Deferred**: the `internal/commands/builtin` package is a sub-package of `internal/commands`; the `commands.tr()` seam is unexported and not reachable from `builtin`, and `Description()` / `Usage()` are context-free synchronous methods. Migrating this surface requires a dedicated seam refactor (own `builtin/i18n` package + context plumbing) — scoped to a future round to keep round-350 focused and low-risk.

Round-350 migration

`server/handlers.go`: 16 distinct user-facing JSON `"message"` literals migrated to the existing `internal/server` i18n seam (`internal/server/i18n_seam.go` + `i18n/bundles/active.en.yaml`), covering ~73 call sites. Six bundle IDs from round-177 that were declared but never wired into handler code are now wired; ten new IDs added. A nil-safe `reqCtx(c)` helper was added to the seam, which also repairs a pre-existing latent nil-`Request` panic in `qa_handlers.go` (round-70 wiring assumed a non-nil `c.Request`; `gin.CreateTestContext` leaves it nil). Paired-mutation tests added in `i18n_seam_test.go`.

Round-462 cmd+internal final sweep

Final confirm-or-NOOP sweep of `helix_code/cmd/` + `helix_code/internal/` (2026-05-20). One genuine (C) residual was located and migrated; the remainder of the audit surface in both trees is confirmed out-of-scope.

Migrated — `cmd/local_llm.go`: the five *advanced* discovery/analytics cobra subcommands (`discoverCmd`, `recommendCmd`, `analyticsCmd`, `reportCmd`, `insightsCmd`) carried plain `Short:` / `Long:` English literals — 10 literals total. These are `--help` text genuinely surfaced to operators and were the only `Short:/Long:` literals in all of `helix_code/cmd/*.go` not already routed through `trc()`. Migrated to the `cmd` i18n seam (`cmd/i18n_seam.go`): 10 new bundle IDs in `cmd/i18n/bundles/active.en.yaml` (`cmd_local_llm_{discover,recommend,analytics,report,insights}_{short,long}`). Paired-mutation tests added to `cmd/local_llm_i18n_test.go` (`round462CobraMetadataIDs` + `TestLocalLLMI18n_Round462CobraMetadata{RoutesThroughSeam,NoopEcho}`).

Confirmed exhausted — out-of-scope residual:

- `cmd/local_llm.go:580` — `"watch: fsnotify error: %v\n"`: class (B) wrapped-error / log diagnostic.
- `cmd/local_llm_advanced.go` — `"Llama 3 8B Instruct"`, `"Mistral 7B Instruct"`, etc.: model-identifier metadata tokens, not localizable UI; `"Total Models\t%d\n"` etc.: tab-separated report table fragments (class B).
- `cmd/performance_optimization/main.go`, `cmd/security_fix_standalone/main.go`, `cmd/security_scan/main.go` — recommendation-record / scanner-finding struct fields + report-template strings (round-460 documented these as out-of-scope: scanner-diagnostic struct fields + report templates).
- `cmd/cli/main.go` — pprof diagnostic strings (class B).
- `internal/*` — the top packages (`tools`, `llm`, `workflow`, `agent`, `memory`, `performance`, `server`, etc.) were classified A/B in revision 1; `agent/base_agent.go` lines 525-722 are class (A) LLM prompt templates; `adapters/speckit_debate_adapter/adapter.go` markdown report fragments are class (B). The genuine (C) surface (`server/handlers.go` JSON `message` fields, `commands/builtin/* Description()/Usage()`) was migrated in earlier rounds — the `internal/commands/builtin` package now has its own `builtin/i18n` + `builtin/translator.go` seam and every `Description()/Usage()` returns `trc("builtin_*_...", nil)`.

Verdict: `helix_code/cmd/` and `helix_code/internal/` genuine user-facing (C) hardcoded-content surface is now exhausted. Remaining audit hits are all class (A) LLM prompts, class (B) wrapped-error / log / report-template strings, or identifier/format-spec tokens.

Precedent

- Round-321 — HelixLLM `openai.go`: LLM prompt templates ruled OUT of CONST-046 scope.
- Round-326 — LLMsVerifier `verifier.go`: same ruling reaffirmed.
- Round-177 — `internal/server` i18n seam established (round-70 of the CONST-046 Phase-4 numbering).

- Round-462 — `cmd/local_llm.go` advanced-subcommand cobra Short/Long migrated; cmd+internal genuine (C) surface confirmed exhausted.